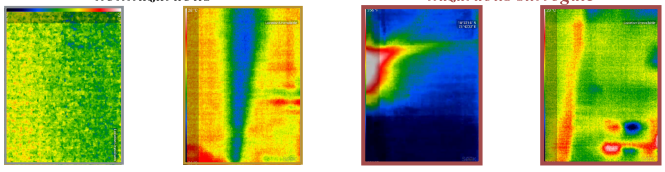
## The corpus, and the action each class maps to

*nonhazardous* — *hazardous surrogate*

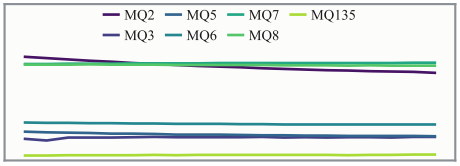


**NoGas**
clean air
a = 0
Monitor

**Perfume**
deodorant vapor
a = 1
Increase sampling

**Smoke**
incense smoke
a = 3
Raise alarm

**Mixture**
smoke and vapor
a = 4
Recommend ESD




**Seven-channel MOX window, 20 × 7**
the only evidence the decision component receives
the thermal path adds no term to the decision state

*no training window shares a raw row with a test window*

## One protocol, and every comparator trained under it

Windowing
length 20
boundary windows dropped

Block-wise holdout
contiguous 20% per class
20-window embargo
seed-varying position

4,900 train / 1,264 test
316 windows per class

| | | | | |
|---|---|---|---|---|
| Cost-weighted policy | Unweighted network | Gradient boosting | Random forest | Recurrent (LSTM) |
| Conservative Q-learning | Perceptron | k-nearest neighbors | Support-vector classifier | Shallow tree depth 3 |

Auxiliary reference, outside the benchmark: CUSUM (two-action)

*the same comparators under one protocol*

| In distribution | Class exclusion | Graded degradation | Raw-space faults | Closed-set reject |
|---|---|---|---|---|
| 30 randomized runs | one hazardous class withheld | noise, drift, channel loss | stuck, zero, clamp, noise | confidence threshold |
| **Decision accuracy** | **Missed hazard** | **Escalation adequacy** | **Under-escalation** | **High-severity false alarm** |

*alarm burden (Eq. 9) is the high-severity false-alarm rate in operational units, not a sixth quantity*